

Nandini Chowdary Muddana

Data Engineer

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SUMMARY

Data Engineer with 4+ years of hands-on experience designing, building, and optimizing large-scale data pipelines and analytics platforms for enterprise clients. Proven expertise in Python, SQL, Apache Spark, cloud data services, and modern data warehousing solutions. Strong background in batch and real-time data processing, data modeling, ETL orchestration, and performance optimization. Adept at transforming raw, high-volume data into trusted, analytics-ready datasets that support business intelligence, machine learning, and executive reporting. Known for delivering scalable, cost-efficient, and highly reliable data solutions aligned with USA enterprise standards.

PROFESSIONAL EXPERIENCE

Data Engineer IBM USA	Oct 2024 – Current
- Designed and implemented end-to-end batch and near real-time data pipelines using Python, SQL, and Apache Spark to process multi-terabyte datasets from transactional systems, APIs, and log sources, improving data availability for analytics teams by 40% and reducing manual data preparation effort by over 55% across enterprise reporting use cases. - Built scalable cloud-native data ingestion and transformation workflows using AWS services including S3, Glue, EC2, and Redshift, enabling reliable ingestion of structured and semi-structured data while achieving 99.9% pipeline uptime and reducing infrastructure costs by approximately 28% through optimized resource utilization. - Developed and optimized complex Spark and Spark SQL jobs with advanced partitioning, caching, and join strategies, significantly improving data processing performance and reducing average job execution time from 4.5 hours to under 2 hours for critical business datasets. - Implemented robust data modeling solutions using star and snowflake schemas in Snowflake and Redshift, creating analytics-ready fact and dimension tables that improved BI dashboard query performance by 45% and supported self-service analytics for business stakeholders. - Orchestrated end-to-end ETL workflows using Apache Airflow, designing DAGs with dependency management, failure handling, and alerting, which reduced pipeline failure resolution time by 60% and improved overall data pipeline reliability. - Collaborated closely with data analysts, data scientists, and business stakeholders to translate reporting and analytics requirements into scalable data engineering solutions, directly supporting KPI dashboards, forecasting models, and executive-level decision-making. - Implemented automated data quality checks, schema validations, and reconciliation processes within ETL pipelines, reducing data inconsistencies by 35% and increasing stakeholder trust in enterprise data assets. - Leveraged CI/CD practices and Git-based version control to automate deployment of data pipelines and configuration changes, reducing deployment errors by 30% and accelerating release cycles for new data features. - Monitored pipeline performance and data freshness using logging, metrics, and alerting mechanisms, proactively identifying bottlenecks and ensuring SLA compliance for business-critical datasets.	

Data Engineer Cognizant India	Jan 2021 – Dec 2023
- Designed and developed large-scale ETL pipelines using Python, SQL, and Apache Spark to ingest, transform, and load high-volume data from multiple source systems, enabling downstream analytics and reporting for global enterprise clients and improving data processing throughput by 38%. - Built and maintained cloud-based data lakes using AWS S3 as the central storage layer, implementing structured folder strategies and metadata management that improved data discoverability and reduced data retrieval time by 30%. - Developed complex SQL transformations and optimization strategies across relational and cloud data warehouses, reducing query execution times by up to 45% and improving overall reporting responsiveness for business users. - Implemented batch data processing workflows using AWS Glue and custom Python frameworks, ensuring reliable data ingestion schedules and achieving consistent data delivery for daily and hourly reporting pipelines. - Designed dimensional data models aligned with business reporting requirements, enabling accurate KPI tracking and historical trend analysis across finance, operations, and customer analytics domains. - Collaborated with cross-functional teams including QA, DevOps, and business analysts to ensure data accuracy, completeness, and consistency throughout the data lifecycle, reducing production data issues by 32%. - Applied data validation rules, error handling, and logging mechanisms across pipelines, significantly improving data reliability and reducing manual reprocessing efforts for failed jobs. - Supported performance tuning and cost optimization initiatives by analyzing workload patterns and optimizing compute usage, contributing to a 25% reduction in overall cloud data processing costs. - Documented end-to-end data architectures, ETL logic, and operational procedures, improving knowledge transfer and reducing onboarding time for new data engineering team members by 40%.	

TECHNICAL SKILLS

Programming Languages: Python, SQL, PySpark, Shell Scripting Big Data Frameworks: Apache Spark, Spark SQL, Hadoop HDFS Cloud Platforms: AWS (S3, EC2, Glue, Lambda, Redshift), Azure (ADF, ADLS – working knowledge) Data Warehousing: Snowflake, Amazon Redshift, Data Lake Architecture ETL & Orchestration: Apache Airflow, AWS Glue, Custom Python ETL Frameworks Streaming & Messaging: Kafka fundamentals, Event-driven data ingestion patterns Databases: PostgreSQL, MySQL, NoSQL concepts (DynamoDB basics) Data Modeling: Star Schema, Snowflake Schema, Dimensional Modeling, Fact & Dimension Design Data Quality & Governance: Data validation checks, schema enforcement, data reconciliation, lineage tracking DevOps & Version Control: Git, CI/CD pipelines, automated deployments Performance Optimization: Query tuning, Spark job optimization, partitioning strategies Analytics Enablement: BI data readiness, KPI datasets, reporting layer optimization	
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EDUCATION

Master of Computer Science and Information Technology Sacred Heart University, Connecticut, USA	Jan 2024 – Mar 2025
Bachelor of Computer Science Vignan's Foundation For science and technology, Andhra Pradesh, India	July 2019 – May 2023